

AARHUS UNIVERSITY

DEPARTMENT OF MATHEMATICS

INTRODUCTION TO ALGEBRA 1

Handin

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September 7, 2026



Goofy ah Math Number 2

Determine $\gcd(175, 63)$.

k	r_k	q_k	a_k	b_k
0	175	-	1	0
1	63	-	0	1
2	49	2	1	-2
3	14	1	-1	3
4	7	3	4	-11

Table 1: Extended Euclidean Algorithm finding $\gcd(175, 63)$

Determine the integers x and y so $7 = 175 \cdot x + 63 \cdot y$

k	r_k	q_k	x_k	y_k
0	175	-	1	0
1	63	-	0	1
2	49	2	1	-2
3	14	1	-1	3
4	7	3	4	-11

Table 2: Extended Euclidean Algorithm finding $\gcd(175, 63)$

Convince God that there doesn't exist some integer r and s so
 $50 = 175 \cdot r + 63 \cdot s$